

BBH branch

BBH free data (Binary.hpp)
superpose 2× WrappedGr(KerrSchild)
 $K \neq 0$; $\bar{u} = 0$ (quasi-equilibrium)
 $\beta_{\text{bg}} = \Omega \times r + \dot{a} r$

BBH domain + BCs
BinaryCompactObject: 2 excisions,
envelope, outer shell $R=10^9$
ApparentHorizon ($\Theta=0$) + outer Flatness

BBH control loop
PostprocessId/ControlId.py:
mass, spin, CoM, $P_{\text{ADM}} \rightarrow$ re-solve

shared spine

free data
 $\bar{\gamma}_{ij}, \bar{u}_{ij}, K, \partial_t K$, sources
system::background_fields

XCTS elliptic system
Xcts::FirstOrderSystem<
HamiltonianLapseAndShift,Curved,0>
solve $\psi, \alpha\psi, \beta_{\text{exc}}^i$

DG discretization
WeakInertial, Massive, GaussLobatto

NewtonRaphson
globalized nonlinear solve

linear-solver stack
Gmres + Multigrid + Schwarz
(MinusLaplacian / ExplicitInverse)

ADM reconstruction
SpacetimeQuantitiesCompute:
 $\gamma_{ij} = \psi^4 \bar{\gamma}_{ij}, K_{ij}, \alpha, \beta^i, H, M$

observers
field_observations + ObserveNorms
+ ObserveAdmIntegrals

Reductions H5
NewtonRaphson residuals,
 $\|H\|_2, \|M\|_2$

Volume H5
 ψ, α, β^i ,
SpatialMetric, ExtrinsicCurvature

Teukolsky branch

Teukolsky free data
(TeukolskyWave.hpp)
 $\bar{\gamma}_{ij} = \delta_{ij} + h_{ij}$ ($\ell=2, m=0$)
 $K=0, \partial_t K=0$; $\beta_{\text{bg}} = 0$
 $\bar{u}^{ij} = -\bar{\gamma}^{ik} \bar{\gamma}^{jl} \bar{h}_{kl}$

Teukolsky domain + BCs
Sphere: regular center,
InnerR 0.5, OuterR 10^9 ,
partition [12, 40, 120]; outer Robin

no control loop
fixed analytic free data;
single XCTS solve

(1) **author/choose free-data class**
elliptic::analytic_data::AnalyticSolution
Binary / TeukolskyWave / DuckTovStar

~20 variables() overloads
ConformalMetric, Inverse, ∂ ;
 K_{ij} , K , $\partial_i K$, $\partial_t K$;
 ψ , $\alpha\psi$, β_{bg} , $\bar{u}$ -term, ShiftExcess;
matter sources (ρ , S , S^i)

(2) **register in executable + CMake**
SolveXcts (hydro obs)
vs SolveXctsVacuum (no hydro)
factory_creation + target_link XctsAnalyticData

(3) **write YAML**
Background, InitialGuess, DomainCreator,
NonlinearSolver/LinearSolver, BCs, Observers

(4) **build**
cmake -build -target SolveXctsVacuum

(5) **run**
./SolveXctsVacuum -input-file ... +pN

(6a) **Reductions H5**
convergence + $\|H\|_2, \|M\|_2$

(6b) **Volume H5**
ConformalFactor, Lapse, Shift,
SpatialMetric, ExtrinsicCurvature

(7) **consume** → **AthenaK**
spectre::Exporter::interpolate_to_points<3>
→ z4c_spectre_bbh.cpp::LoadSpectreInitialData
SpatialMetric_xx.zz, ExtrinsicCurvature_xx.zz
→ ADMToZ4c<N> → Z4c evolution